

# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

## INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

### LAB 07 – EXPLORING ICS PACKET CAPTURES

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**Course:** Intro to Critical Infrastructure for IT Learners  
**Tool:** Wireshark  
**Capture:** ICDFA\_Lab7ICS\_Traffic.pcap  
**Date:** 2 October 2026

## Objective

The objective of this lab was to use Wireshark to analyze an instructor- provided ICS packet capture and identify HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7 communication, while documenting security-relevant observations. :chatgpt-content-referenceindex="1"

## 1. HTTP Basic Authentication

**Filter:** http.authbasic

HTTP Basic Authentication material was visible in the capture. The credential value has been redacted from the submitted evidence as required by the laboratory instructions.

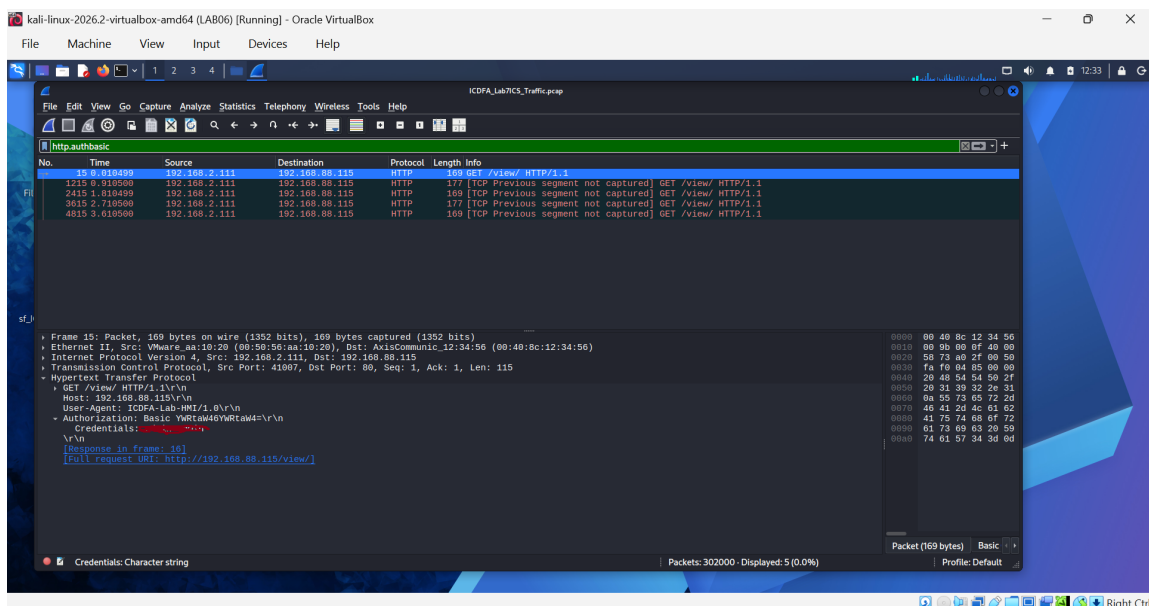


Figure 1: FTP server metadata

## 2. HTTP POST Requests

Filter: `http.request.method == "POST"`

The capture contained POST requests including authentication-related form submissions. Sensitive values have been redacted.

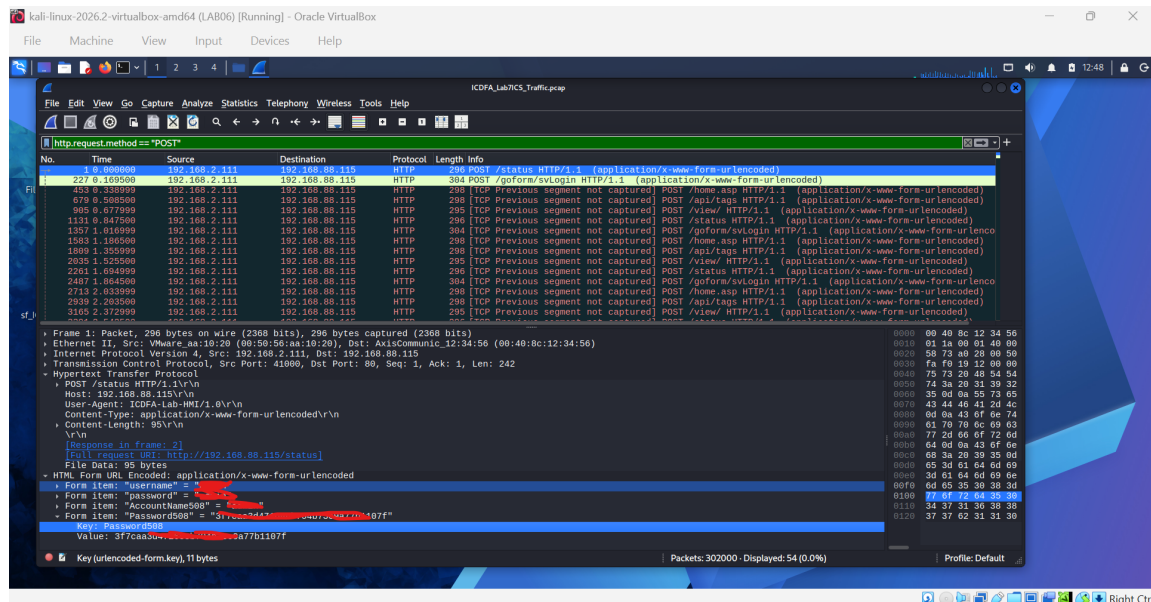


Figure 2: FTP server metadata

## 3. FTP

Filter: `ftp`

The FTP server banner identified an:

- AXIS 206 Network Camera
- FTP Server version 4.40
- Response code 220

This demonstrates that FTP traffic can expose device and software metadata.

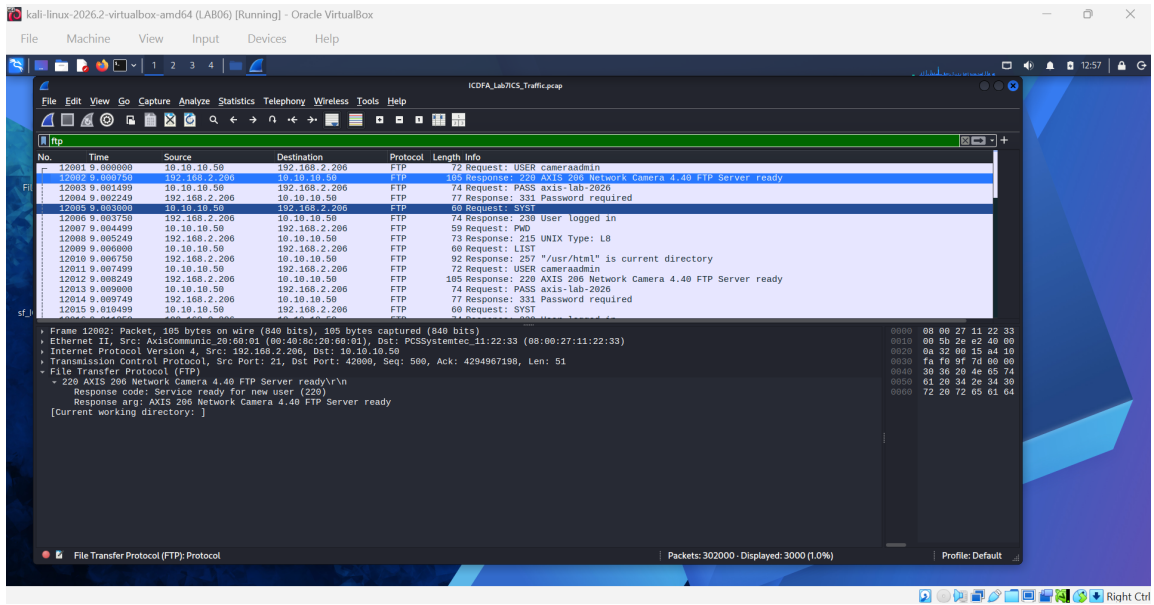


Figure 3: FTP server metadata.

## 4. Modbus

Filter: modbus

The capture contained several Modbus function codes:

- Function Code 1 – Read Coils
- Function Code 3 – Read Holding Registers
- Function Code 6 – Write Single Register
- Function Code 43 – Read Device Identification

The selected packet demonstrates Function Code 1, Read Coils.

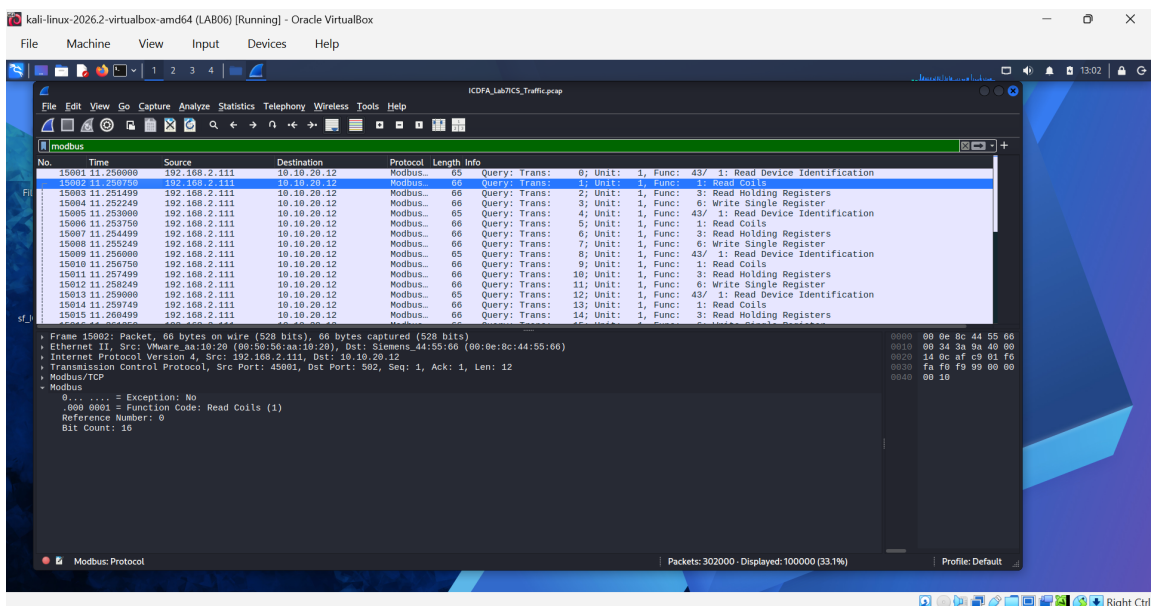


Figure 4: Modbus Function Code 1 – Read Coils.

## 5. DNP3

Filter: dnp3

The dnp3 filter returned no decoded packets in the supplied capture. Further inspection using:

```
tcp.port == 20000
```

showed traffic from 10.10.10.50 to 10.10.30.20. The selected packet contained TCP destination port 20000 and payload beginning with 05 64. Wireshark did not decode this traffic as DNP3, even after enabling the DNP3 TCP dissector, so no unsupported DNP3 application-layer fields are claimed.

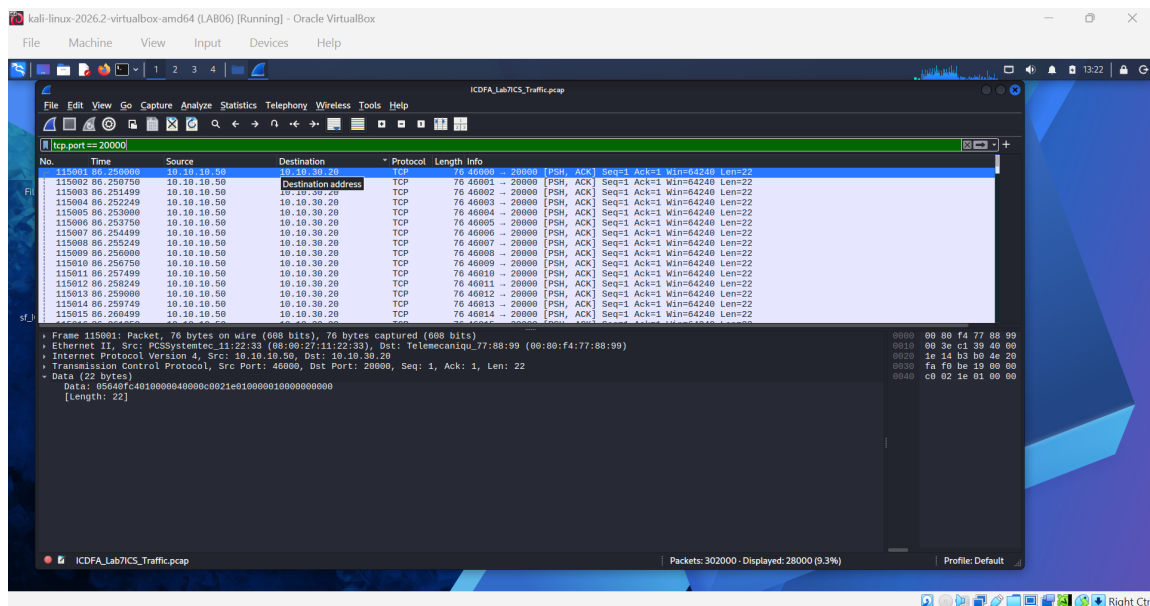


Figure 5: TCP port 20000 traffic with payload beginning with 05 64.

**Security relevance:** DNP3 is commonly used in SCADA and process automation environments for communication between control systems and field devices.

## 6. EtherNet/IP

Filter: enip

The selected packet showed an EtherNet/IP Register Session request.

- Source: 10.10.10.50
- Destination: 10.10.20.11
- Command: Register Session (0x0065)
- Status: Success
- Protocol Version: 1

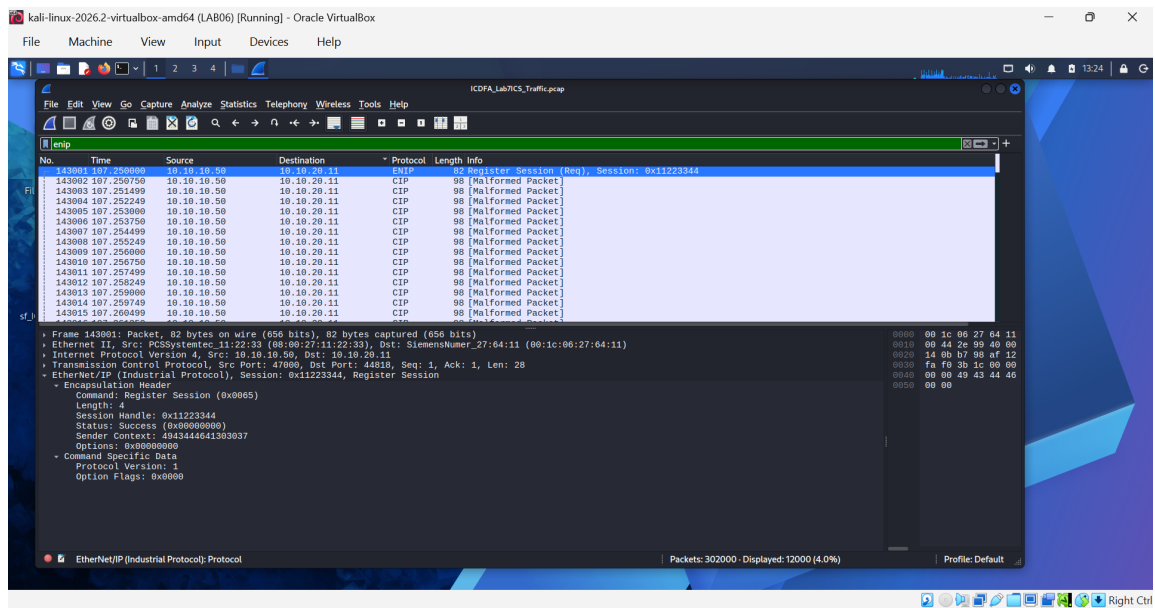


Figure 6: EtherNet/IP Register Session.

## 7. S7 Communication

Filter: s7comm

The selected packet showed an S7 Read Var operation.

- Source: 10.10.10.50
- Destination: 10.10.20.11
- TCP Port: 102
- Function: Read Var (0x04)
- Item Count: 1
- Area: Data Blocks (DB)

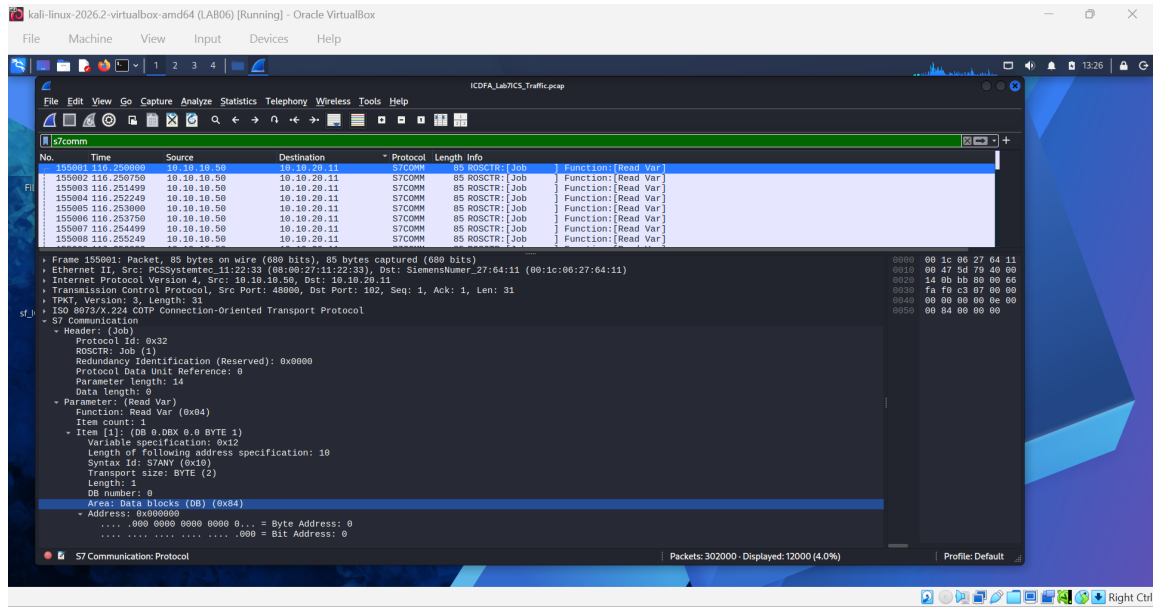


Figure 7: S7 Communication – Read Var operation.

## 8. Protocol Evidence Matrix

| Protocol        | Filter                                     | Evidence / Observation                                                                     |
|-----------------|--------------------------------------------|--------------------------------------------------------------------------------------------|
| HTTP            | <code>http.authbasic</code>                | Basic Authentication material visible in HTTP Authorization field. Credential redacted.    |
| HTTP POST       | <code>http.request.method == "POST"</code> | Authentication-related form fields were visible in POST traffic.                           |
| FTP             | <code>ftp</code>                           | AXIS 206 Network Camera, FTP Server 4.40, response 220.                                    |
| Modbus          | <code>modbus</code>                        | Function Codes 1, 3, 6 and 43 observed; Read Coils selected for evidence.                  |
| DNP3            | <code>dnp3 / tcp.port == 20000</code>      | TCP/20000 traffic observed with payload beginning 05 64; not decoded by Wireshark as DNP3. |
| EtherNet/IPenip |                                            | Register Session request, command 0x0065, status Success.                                  |
| S7              | <code>s7comm</code>                        | Read Var operation, source/destination and Data Block information visible.                 |

## 9. Security Observations

The packet analysis revealed the following security-relevant observations:

1. HTTP Basic Authentication material was visible in the captured traffic.
2. HTTP POST traffic exposed authentication-related form fields.
3. FTP exposed device and software-version information through its server banner.

4. Modbus traffic exposed industrial read/write operations.
5. S7 traffic exposed a Read Var operation and addressed data-area information.
6. EtherNet/IP traffic exposed industrial protocol and session information.
7. Multiple industrial protocols were visible in the same capture, demonstrating the amount of information that can be obtained through passive packet analysis.

**Recommended controls:** network segmentation, firewall access controls, least privilege, secure management channels, protocol-aware monitoring, intrusion detection, asset inventory and appropriate protection of authentication traffic.

## 10. Knowledge Check

### 1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is encoding, not encryption. It can be decoded easily and therefore does not protect authentication information from someone who captures the traffic.

### 2. What information can device banners or protocol metadata reveal?

They can reveal device type, vendor, model, software version, protocol, services, endpoints and information about the operations being performed.

### 3. How do Modbus, DNP3, EtherNet/IP and S7 differ from HTTP or FTP?

They are industrial communication protocols designed for communication between control systems, PLCs and field devices. HTTP and FTP are general-purpose application protocols commonly used for web and file-transfer services.

### 4. Why is passive packet analysis valuable for understanding an ICS environment?

It allows an analyst to identify devices, protocols, communication paths and operations without actively interacting with potentially sensitive industrial systems.

### 5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Network segmentation, firewall rules, access control, monitoring, intrusion detection, secure management channels, least privilege and asset inventory can reduce exposure.

## Conclusion

The laboratory demonstrated how Wireshark can be used to identify and analyze ICS and conventional network protocols from recorded traffic. The analysis showed that

authentication information, device metadata and industrial control operations may be visible through passive packet inspection, highlighting the importance of segmentation, monitoring and appropriate security controls in ICS environments.